2021

AP® Microeconomics

Scoring Guidelines
Set 2

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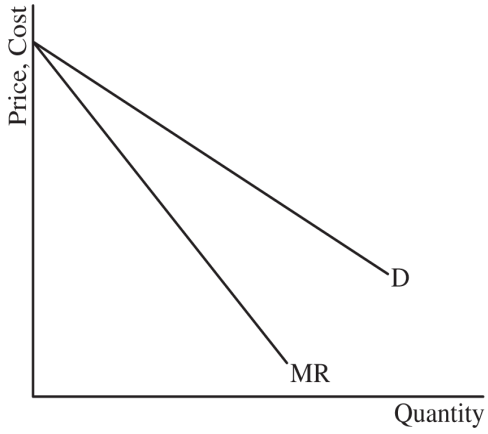
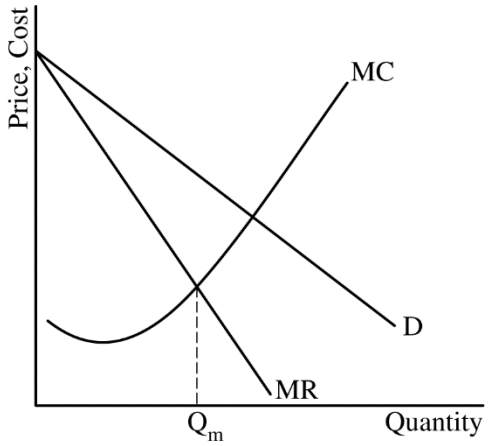
Question 1: Long — 10 points

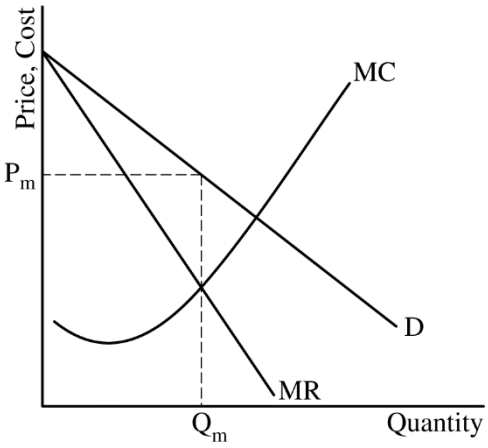
(a) Draw a correctly labeled graph for NCHart showing downward-sloping demand (D) and marginal revenue (MR) curves with the marginal revenue curve below the demand curve. **1 point**



For the second point, the graph must show the marginal cost (MC) curve and the profit-maximizing quantity, labeled Q_m, where MR=MC. **1 point**



For the third point, the graph must show the profit-maximizing price, labeled P_m, above Q_m from the demand curve. **1 point**



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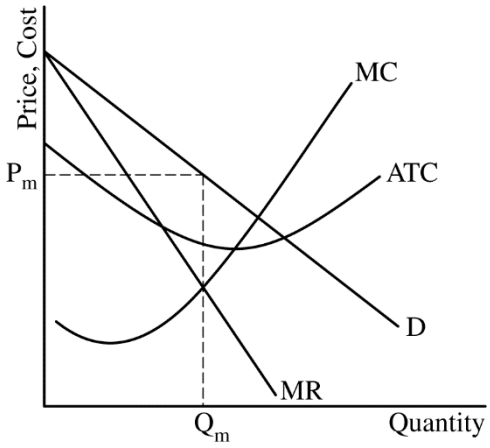
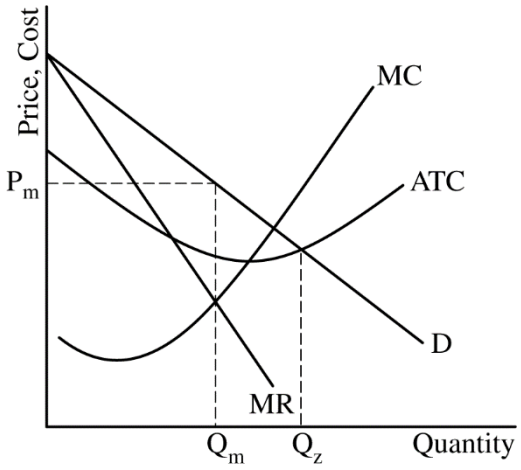
For the fourth point, the graph must show the ATC below the demand curve at Q_m with the MC curve rising and intersecting the ATC curve at its minimum. **1 point**



Total for part (a) 4 points

(b) State that demand is elastic and explain that MR is positive at Q_m or that Q_m is less than the quantity at which marginal revenue equals zero. **1 point**

(c) (i) On your graph from part (a), show the quantity that is consistent with the goal of NCHart generating enough revenue to cover its total costs labeled as Q_Z. **1 point**



(ii) State there is a deadweight loss at Q_z and explain that P (or D)<MC, as shown. **1 point**

Note: Deadweight loss will exist at Q_z if the demand is drawn such that the quantity at which D=ATC is less than the quantity at which D=MC, because P (or D)>MC.

Note: Deadweight loss will NOT exist at Q_z if the demand is drawn such that the quantity at which D=ATC is equal to the quantity at which D=MC, because P (or D)=MC.

Total for part (c) 2 points

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(d) (i)	State that no, TXDrug does not have a dominant strategy, and explain that if NCHart chooses Q_m, then TXDrug's best response is to Enter because \$1 > \$0, but if NCHart chooses Q_Z, then TXDrug's best response is to Stay Out because \$0 > −\$1.	**1 point**
(ii)	State that the best response for NCHart is to produce Q_m.	**1 point**
(iii)	Identify the Nash equilibrium as NCHart produces Q_m and TXDrug chooses to Enter.	**1 point**
	Total for part (d)	**3 points**
	Total for question 1	**10 points**

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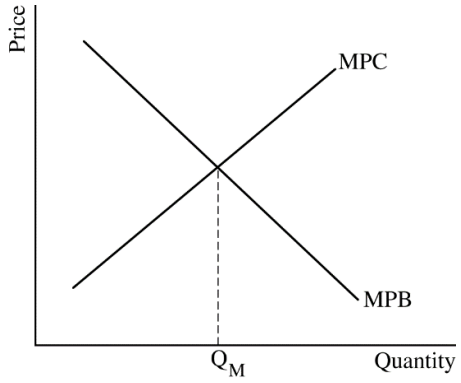
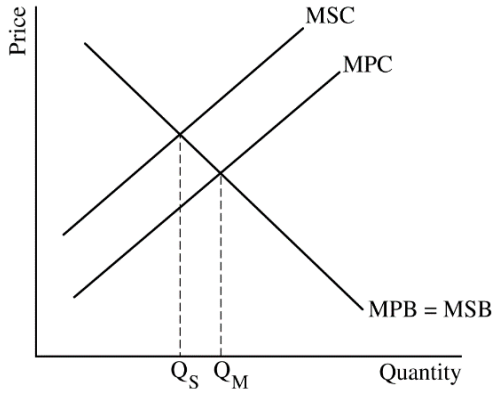
Question 2: Short — 5 points

(a) Draw a correctly labeled graph with an upward-sloping supply curve labeled MPC, a downward-sloping demand curve labeled MPB, and the market equilibrium quantity labeled Q_M. **1 point**



For the second point, the graph must include the MSC curve above the MPC curve at all output levels and must show the socially efficient quantity labeled Q_S. **1 point**



Total for part (a) 2 points

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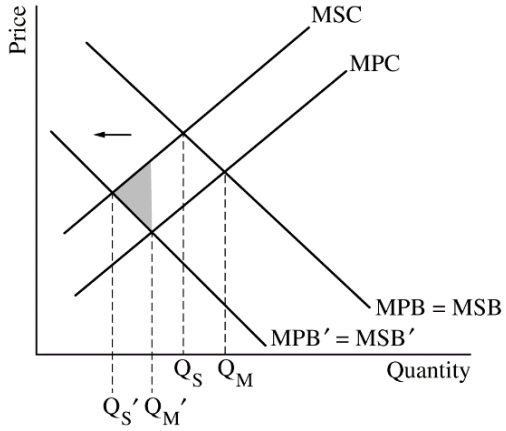
(b)	On your graph from part (a), show a leftward shift of the demand curve and shade completely the area of deadweight loss at the new market equilibrium.	**1 point**



(c) (i)	State that the per-unit tax would be equal to the marginal external cost (MSC − MPC).	**1 point**
(ii)	Explain that the lump-sum tax will not change the quantity produced because it does not affect the marginal cost.	**1 point**
	Total for part (c)	**2 points**
	Total for question 2	**5 points**

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Question 3: Short — 5 points

(a)	Calculate the total net benefit of placing three advertisements as $2,200 and show your work: Total net benefit = $3,000 – $800 = $2,200	**1 point**
(b)	Calculate the marginal net benefit of the third advertisement as $500 and show your work: Marginal net benefit = ($3,000 – $2,200) – ($800 – $500) = $800 – $300 = $500	**1 point**
(c)	Identify the optimal number of advertisements as 4 and explain that the marginal net benefit of the 4th advertisement is positive ($600 – $500 = $100), but the marginal net benefit of the 5th advertisement is negative ($400 – $800 = –$400).	**1 point**
(d)	Identify the optimal number of advertisements as 4.	**1 point**
(e)	State that AZY Foods is operating in a monopolistically competitive market structure.	**1 point**
	Total for question 3	**5 points**

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